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- Hierarchical models
  - Rats example
  - Factory example
- Exchangeability
- Computational aspects
- Additional material
  - Factory extension
  - 8 schools example

Bayesian Statistics and Data Analysis
Lecture 7

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Department of Statistics, Uppsala University

Thanks to Aki Vehtari, Aalto University



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Section 1

Hierarchical models



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Stan tutorial for hierarchical models

- Before Assignment 7, watch Hierarchical Modelling in Stan: Predicting the Premier League
- While watching, connect the example to:
  - partial pooling
  - group-level parameters
  - comparison with non-hierarchical models
  - Stan diagnostics for hierarchical models

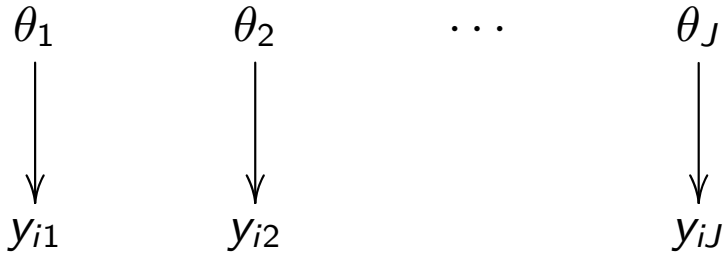


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Survival probabilities across hospitals

- Example: comparing hospitals
  - Hospital j: survival probability θ_j
  - Patient i's survival outcome: y_{ij}



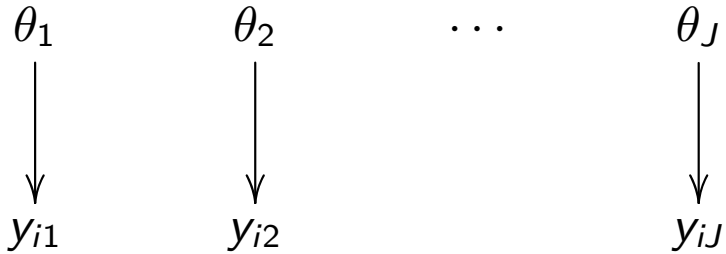


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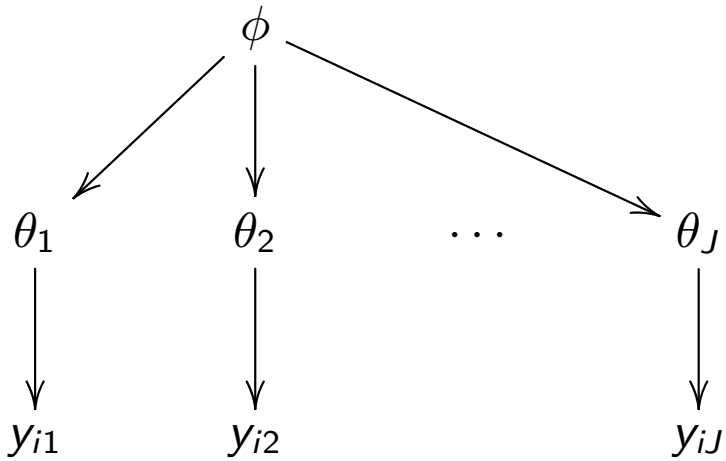
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Survival probabilities across hospitals

- Example: comparing hospitals
  - Hospital j: survival probability θ_j
  - Patient i's survival outcome: y_{ij}



- Model the θ_j as similar, not identical



- A shared population distribution has hyperparameters ϕ
- Both θ_j and ϕ are unknown



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Notation for hierarchical models

- $j = 1, \ldots, J$: groups; $i = 1, \ldots, n_j$: observations in group j
- y_{ij}: individual observation; y_j: group data or a group summary; $y = (y_1, \ldots, y_J)$
- θ_j: group-specific parameter; $\theta = (\theta_1, \ldots, \theta_J)$
- ϕ: population hyperparameters

For normal hierarchies:

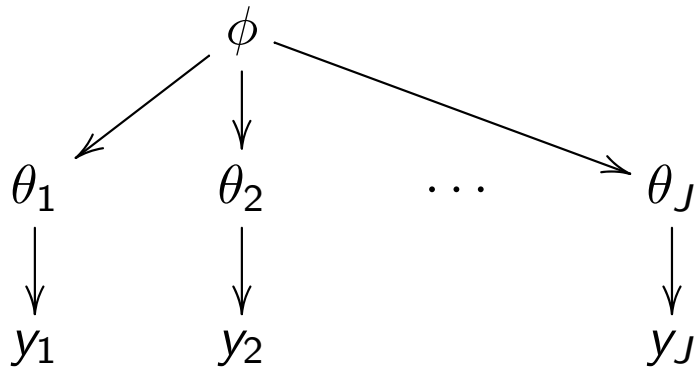
- μ: population mean; τ: between-group SD
- σ: within-group SD of individual observations



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Hierarchical model: assumptions



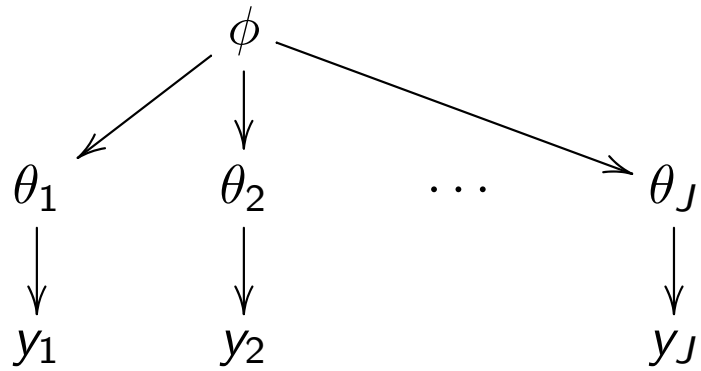
- Given ϕ, the θ_j are independent with a common population distribution $p(\theta_j \mid \phi)$



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Hierarchical model: assumptions



- Given ϕ, the θ_j are independent with a common population distribution $p(\theta_j \mid \phi)$
- Given θ, groups are independent, and y_j depends only on its own parameter θ_j
- The hyperprior $p(\phi)$ describes uncertainty about the population distribution



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Hierarchical model: joint posterior

Bayes' rule and the conditional-independence assumptions give

$$p(\theta, \phi \mid y) \propto p(y \mid \theta)\, p(\theta \mid \phi)\, p(\phi)$$
$$\propto p(\phi) \prod_{j=1}^{J} \left[p(\theta_j \mid \phi)\, p(y_j \mid \theta_j) \right].$$

- One hyperprior; each group contributes a population factor and a likelihood



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$$\propto p(\phi) \prod_{j=1}^{J} \left[p(\theta_j \mid \phi)\, p(y_j \mid \theta_j) \right].$$

- One hyperprior; each group contributes a population factor and a likelihood
- Given ϕ and y, the group parameters are independent
- Averaging over the shared, uncertain ϕ generally makes the θ_j dependent in the posterior
- Each group's data inform ϕ, so other groups can inform its estimate of θ_j

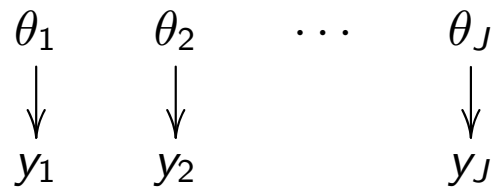


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No, complete, and partial pooling

No pooling (separate)
Independent priors; each θ_j uses only group j's data.



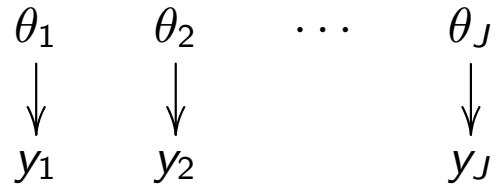


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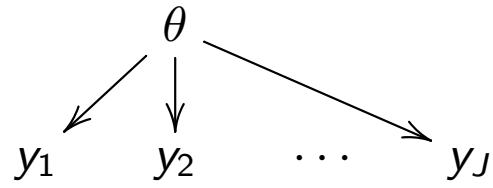
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No, complete, and partial pooling

No pooling (separate)
Independent priors; each θ_j uses only group j's data.



Complete pooling (pooled)
All groups inform one common θ.



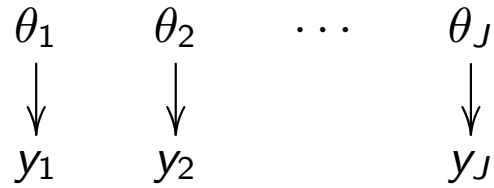


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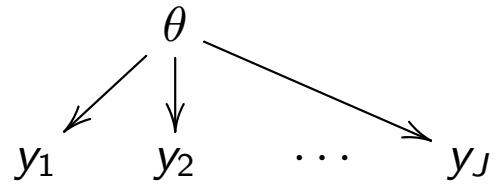
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No, complete, and partial pooling

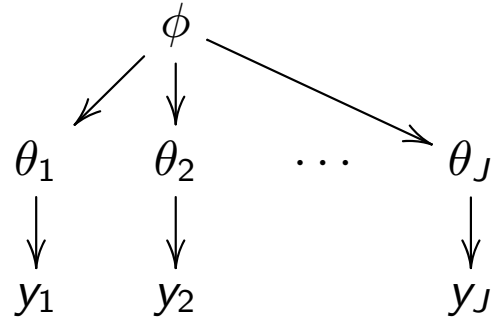
No pooling (separate)
Independent priors; each θ_j uses only group j's data.



Complete pooling (pooled)
All groups inform one common θ.



Partial pooling (hierarchical)
Distinct θ_j share information through ϕ.





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Pair discussion: group size and pooling

Discuss in pairs (1–2 minutes).

Two groups have the same observed success rate:

$$\text{Group A: } \frac{1}{5} = 20\%, \qquad \text{Group B: } \frac{100}{500} = 20\%.$$

Both share the same population information, with a population mean of 32%.

Which estimate should move more toward the population mean? Why?



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Pair discussion: group size and pooling

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Both share the same population information, with a population mean of 32%.

Which estimate should move more toward the population mean? Why?

Answer: Group A. Its five observations provide weaker evidence, so the population information receives more weight.

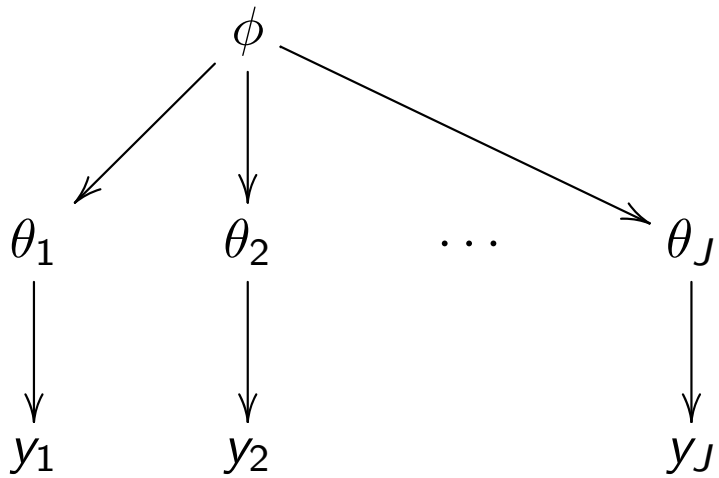


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Predictive distributions for hierarchical models

- Two types of predictive distributions
  1. A new observation in an existing group
  2. A new observation in a new group





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Hierarchical binomial model: rats

- Medicine testing
- Type F344 female rats in control group given placebo
  - count how many get endometrial stromal polyps (benign (noncancerous) growths in the uterus)
  - familiar binomial model example



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Hierarchical binomial model: rats

- Medicine testing
- Type F344 female rats in control group given placebo
  - count how many get endometrial stromal polyps (benign (noncancerous) growths in the uterus)
  - familiar binomial model example
- Experiment has been repeated 71 times

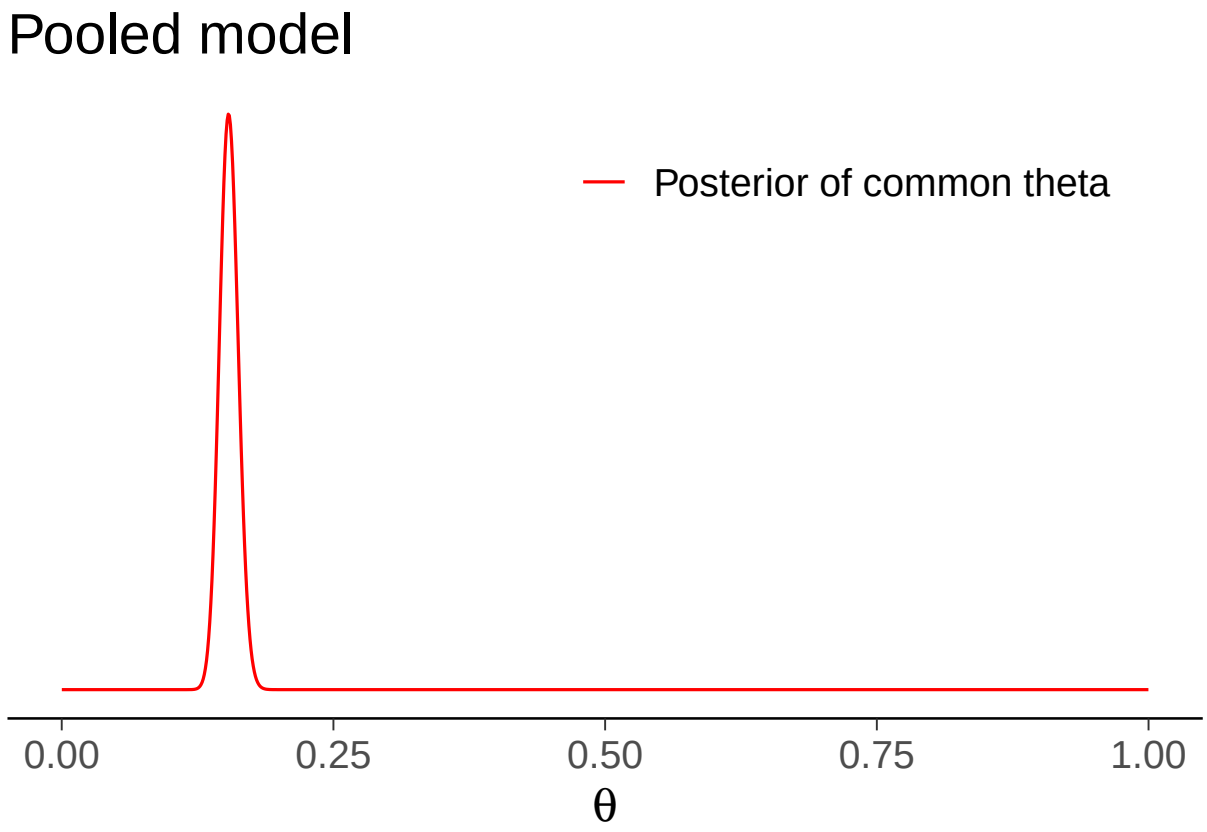
0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/19	0/19	0/19
0/19	0/18	0/18	0/17	1/20	1/20	1/20	1/20	1/19	1/19
1/18	1/18	2/25	2/24	2/23	2/20	2/20	2/20	2/20	2/20
2/20	1/10	5/49	2/19	5/46	3/27	2/17	7/49	7/47	3/20
3/20	2/13	9/48	10/50	4/20	4/20	4/20	4/20	4/20	4/20
4/20	10/48	4/19	4/19	4/19	5/22	11/46	12/49	5/20	5/20
6/23	5/19	6/22	6/20	6/20	6/20	16/52	15/46	15/47	9/24
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Hierarchical binomial model: rats



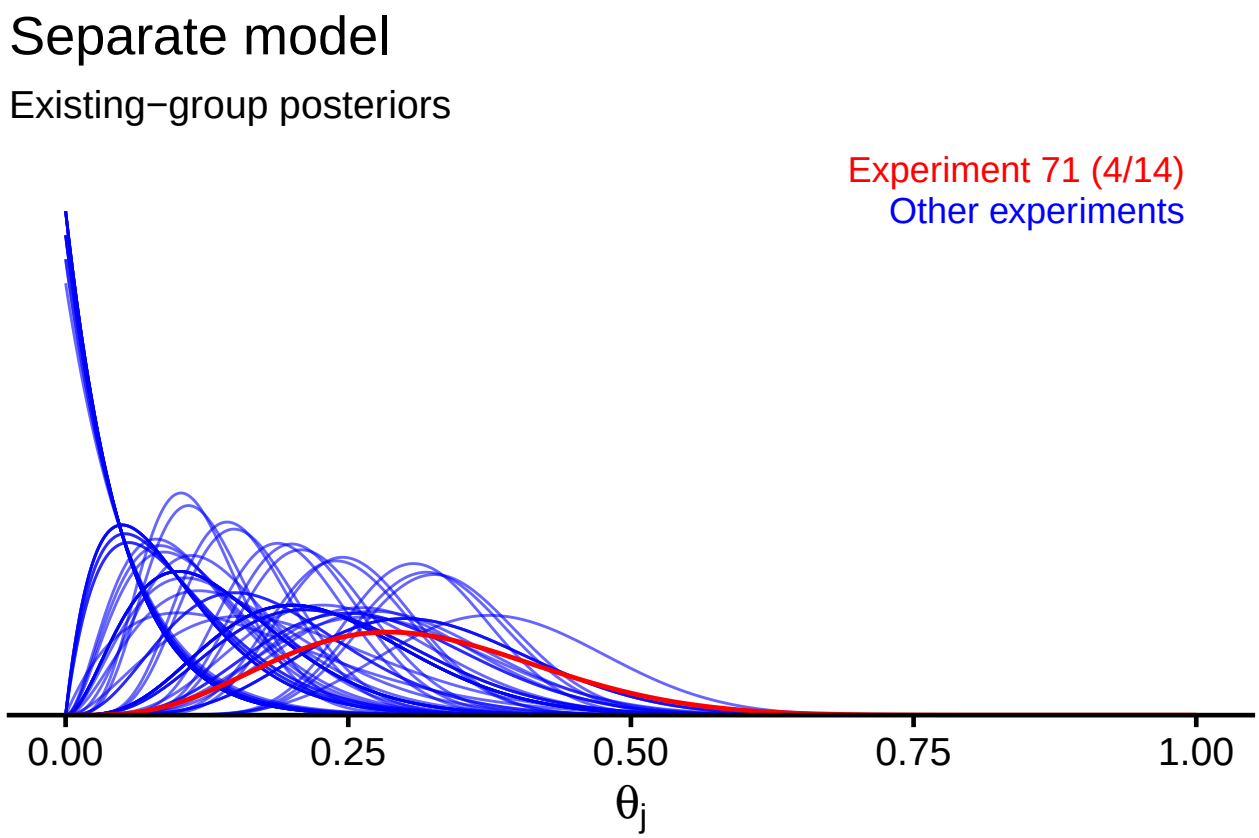
Complete pooling assumes one common probability for all experiments.



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Hierarchical binomial model: rats



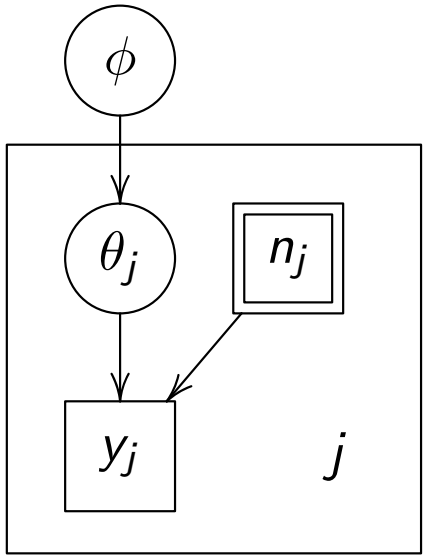
No pooling estimates each probability separately, without learning a shared population distribution.



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Reading the model diagrams



- Circle: unknown quantity
- Single box: observed quantity or a supplied estimate
- Double box: fixed, given quantity
- Arrow: the child's distribution depends on the parent
- Outer box (**plate**): repeat its contents for $j = 1, \ldots, J$

These are the conventions used in this lecture. A nested i plate repeats observations within each group.

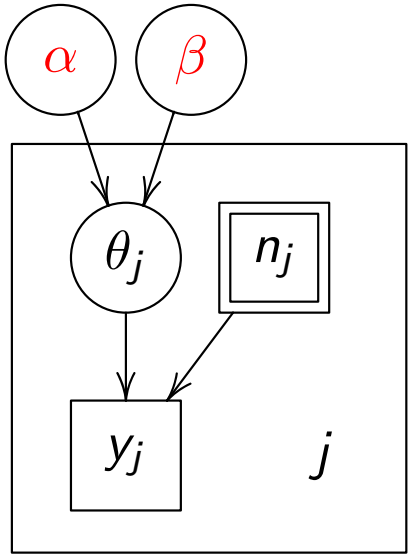


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Hierarchical binomial model: rats

- Here $\phi = (\alpha, \beta)$; y_j counts affected rats out of known n_j



$$\theta_j \mid \alpha, \beta \sim \text{Beta}(\alpha, \beta)$$
$$y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j)$$

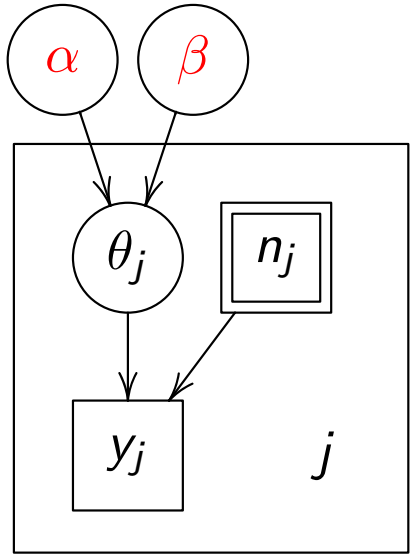


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$$\theta_j \mid \alpha, \beta \sim \text{Beta}(\alpha, \beta)$$
$$y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j)$$

$$p(\theta, \alpha, \beta \mid y) = \left[\prod_{j=1}^{J} p(\theta_j \mid \alpha, \beta, y_j)\right] p(\alpha, \beta \mid y)$$

- One hyperposterior, not one per group
- The θ_j are independent given α, β, y, but generally dependent after averaging over α, β



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Beta-binomial conditional posterior

For fixed $\alpha, \beta > 0$, the binomial likelihood gives

$$\begin{aligned} p(\theta_j \mid \alpha, \beta, y_j) &\propto \theta_j^{y_j}(1-\theta_j)^{n_j-y_j}\theta_j^{\alpha-1}(1-\theta_j)^{\beta-1} \\ &\propto \theta_j^{\alpha+y_j-1}(1-\theta_j)^{\beta+n_j-y_j-1}. \end{aligned}$$

Thus

$$\theta_j \mid \alpha, \beta, y_j \sim \text{Beta}(\alpha + y_j, \beta + n_j - y_j).$$

- The population distribution supplies α, β
- Group j supplies y_j successes and $n_j - y_j$ failures



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Beta-binomial shrinkage

Write the population mean and concentration as

$$m = \frac{\alpha}{\alpha + \beta}, \qquad \kappa = \alpha + \beta.$$

For $n_j > 0$,

$$\begin{aligned} \mathrm{E}(\theta_j \mid \alpha, \beta, y_j) &= \frac{\alpha + y_j}{\alpha + \beta + n_j} \\ &= \underbrace{\frac{n_j}{n_j + \kappa}}_{\text{data weight}} \frac{y_j}{n_j} + \underbrace{\frac{\kappa}{n_j + \kappa}}_{\text{population weight}} m. \end{aligned}$$

- The conditional posterior mean lies between y_j/n_j and m
- With α, β fixed, smaller groups receive more population weight
- Full Bayes also averages over uncertainty in α, β



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Hierarchical binomial model: rats

- Population prior $\text{Beta}(\theta_j|\alpha,\beta)$
- Hyperprior $p(\alpha,\beta)$?
  - α,β both affect the location and scale
  - BDA3 (pp. 110–111) uses, for $\alpha,\beta>0$, $p(\alpha,\beta)\propto(\alpha+\beta)^{-5/2}$
  - This hyperprior is **improper**: its integral is infinite
  - BDA shows that the posterior is proper for these rat data; posterior propriety must be checked when using an improper prior
  - An example-specific choice, not a general default



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Hierarchical inference with posterior draws

For $s = 1, \ldots, S$:

1. Draw $(\alpha^{(s)}, \beta^{(s)}) \sim p(\alpha, \beta \mid y)$
2. For every group j, draw independently conditional on this pair:

$$\theta_j^{(s)} \sim \text{Beta}(\alpha^{(s)} + y_j, \beta^{(s)} + n_j - y_j)$$

Stan note: If θ_j is a model parameter, Stan returns its posterior draws directly.

Share one hyperparameter draw across all groups at each step.

$$\text{E}[h(\theta_j) \mid y] = \int h(\theta_j) p(\theta_j \mid y)\, d\theta_j \approx \frac{1}{S} \sum_{s=1}^{S} h(\theta_j^{(s)}).$$

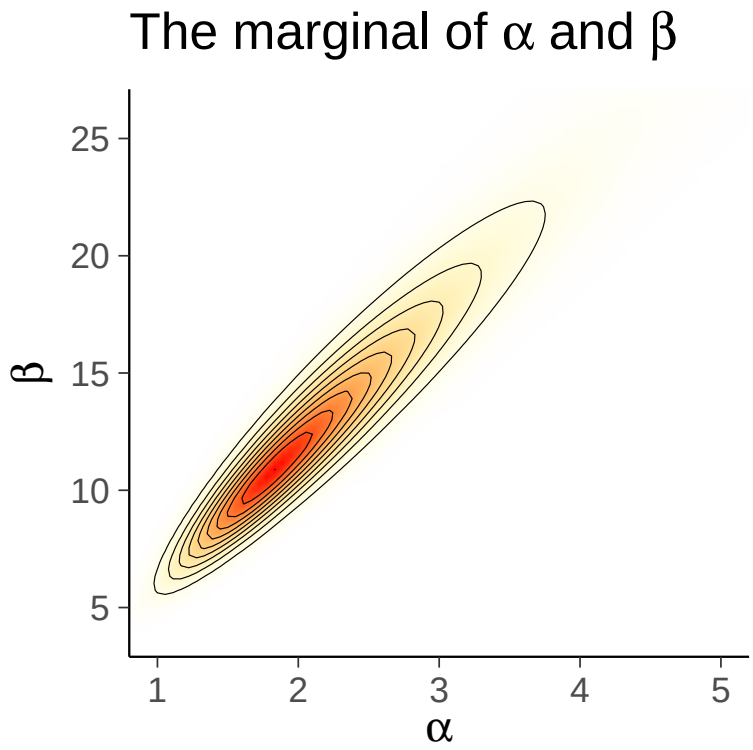
Choosing $h(\theta) = \theta$ estimates a mean; $h(\theta) = \mathbf{1}(\theta > c)$ estimates a probability.



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Rats: learning the population distribution



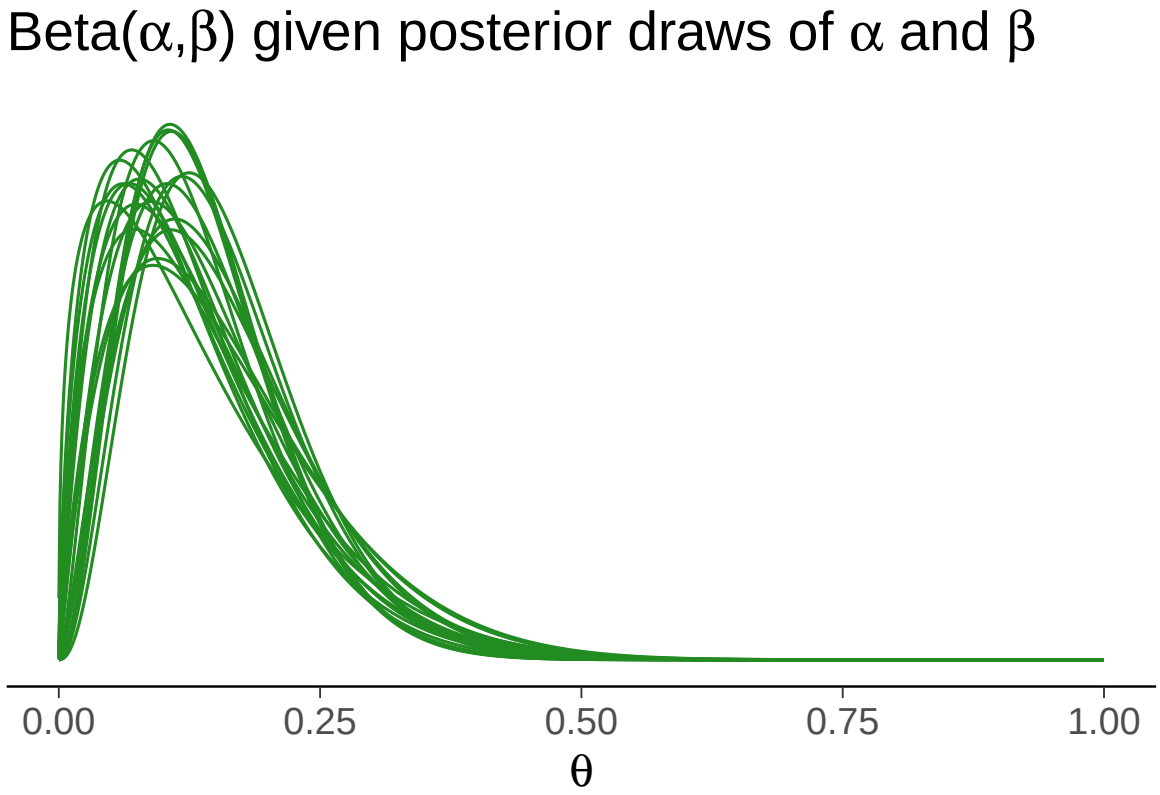
The hyperposterior $p(\alpha, \beta \mid y)$ uses all 71 experiments.



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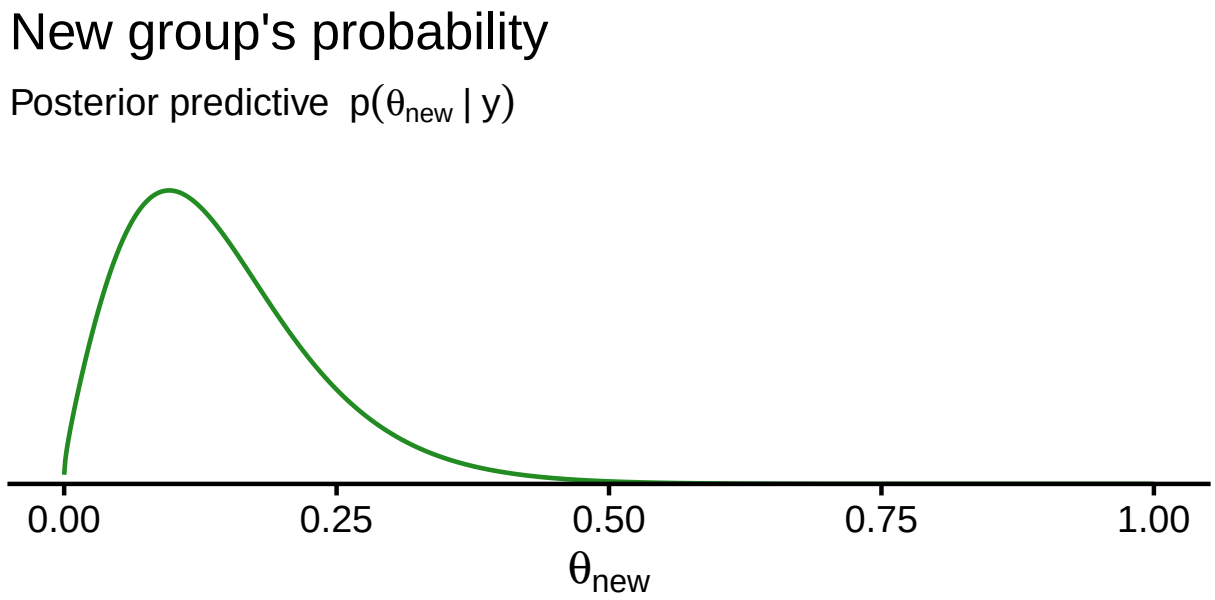
Each curve is Beta(α, β) for one posterior draw of (α, β).



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Rats: a new experiment's probability



$$p(\theta_{\text{new}} \mid y) = \int_0^\infty \int_0^\infty p(\theta_{\text{new}} \mid \alpha, \beta) p(\alpha, \beta \mid y)\, d\alpha\, d\beta.$$

This averages the population curves over the hyperposterior. It predicts a new experiment's probability, before observing its data.

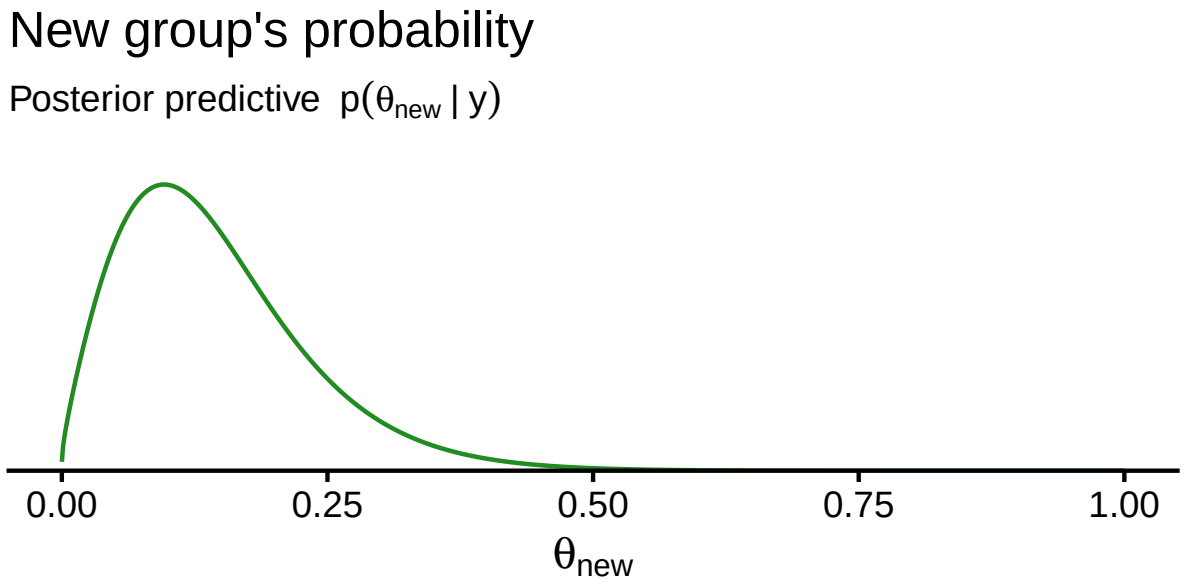


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Pair discussion: a new rat experiment

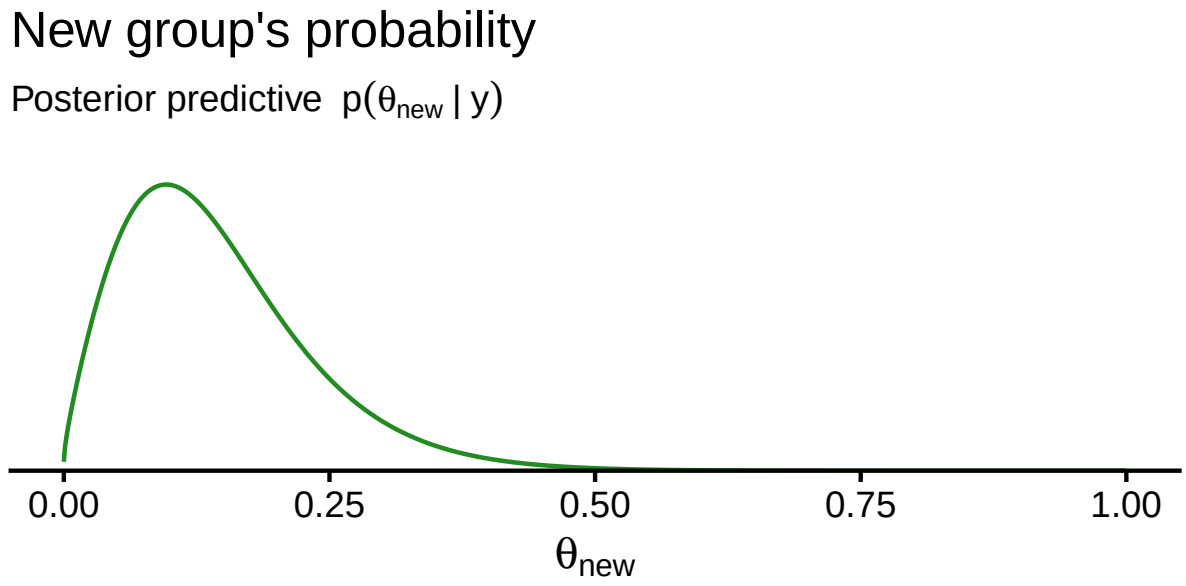
Discuss in pairs (1–2 minutes).



A new experiment will include n_{new} rats.
How would you use this distribution to predict the number of affected rats, y_{new}? Describe the simulation steps.

Pair discussion: a new rat experiment

Discuss in pairs (1–2 minutes).



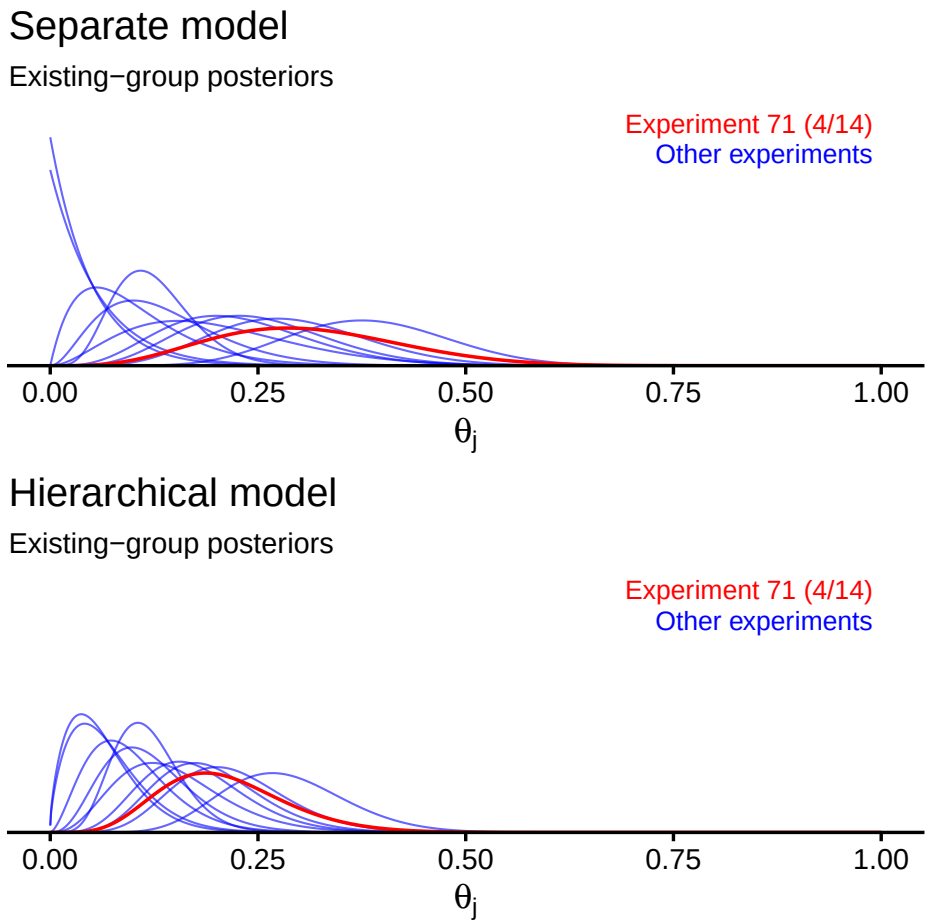
Answer: Draw a probability from the green distribution, then draw a count:

$$\theta_{\text{new}}^{(s)} \sim p(\theta_{\text{new}} \mid y), \qquad y_{\text{new}}^{(s)} \sim \text{Binomial}(n_{\text{new}}, \theta_{\text{new}}^{(s)}).$$

Repeat and summarize the simulated counts.

Rats: partial pooling

Experiment 71 (4/14), shown in red.



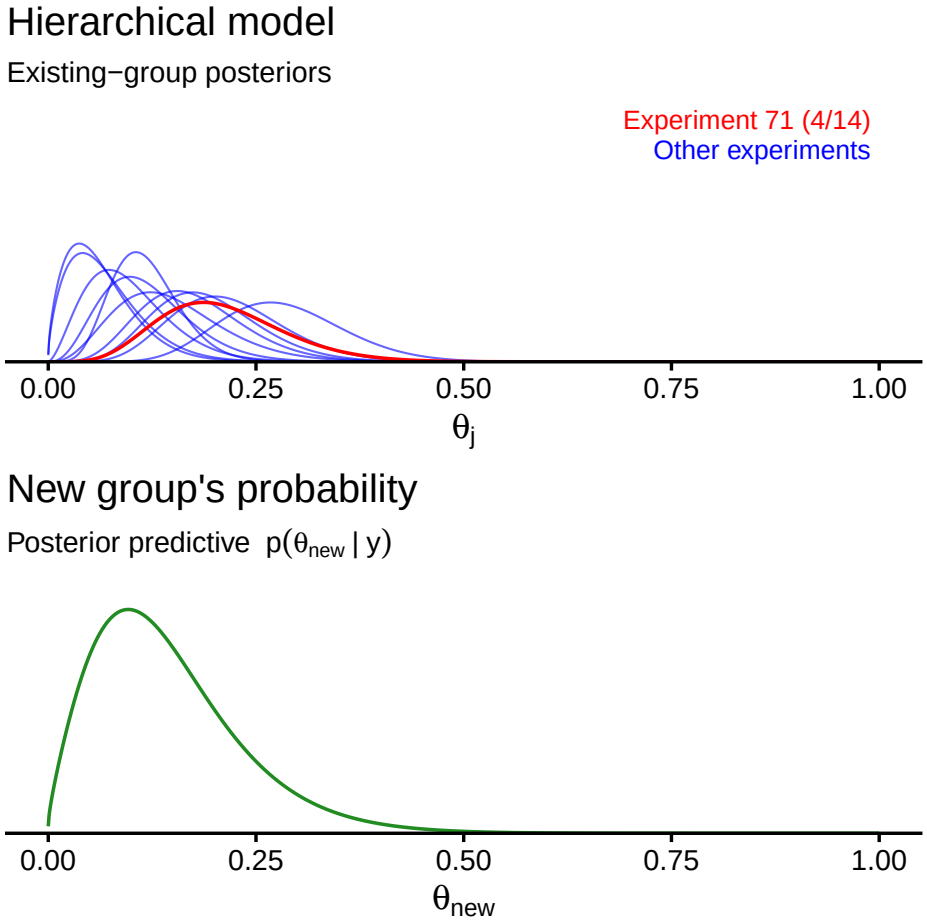
Experiment 71 moves toward the population.
Both plots use the same experiments and scales.



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Rats: existing and new experiments



Existing $p(\theta_j \mid y)$ uses group j's data as well as the population.
New $p(\theta_{\text{new}} \mid y)$ has no new-group data.



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Pair discussion: predictions for rats

Discuss in pairs (1–2 minutes).

What predictive distributions can we have?

Think about both an existing experiment and a new similar experiment.



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Pair discussion: predictions for rats

Discuss in pairs (1–2 minutes).

What predictive distributions can we have?

Think about both an existing experiment and a new similar experiment.

- Future outcomes or counts in an existing experiment, using $p(\theta_j \mid y)$
- A new experiment's probability, $p(\theta_{\text{new}} \mid y)$
- Outcomes or counts in that new experiment, also including binomial sampling variation



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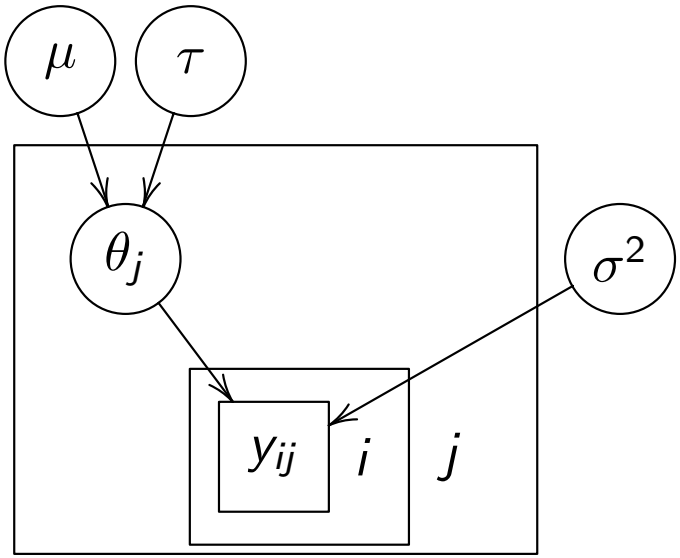
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Hierarchical normal model: factory

- We evaluate the quality produced by six machines in a factory
- Assume hierarchical model
  - each machine has its own (average) quality θ_j and common variance σ^2

$$\theta_j|\mu,\tau \sim \mathcal{N}(\mu,\tau^2)$$

$$y_{ij}|\theta_j,\sigma \sim \mathcal{N}(\theta_j,\sigma^2)$$



- Can be used to predict the future quality produced by each machine and quality produced by a new similar machine



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Factory model: a complete hierarchy

One illustrative choice of **proper, independent priors** is

$$\mu \sim \mathcal{N}(m_0, s_0^2),$$
$$\tau \sim \text{HalfNormal}(0, A_\tau), \qquad \sigma \sim \text{HalfNormal}(0, A_\sigma).$$

The common-variance factory model then has

$$\theta_j \mid \mu, \tau \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \tau^2), \qquad j = 1, \ldots, J,$$
$$y_{ij} \mid \theta_j, \sigma \sim \mathcal{N}(\theta_j, \sigma^2), \qquad i = 1, \ldots, n_j.$$

- HalfNormal$(0, A)$: a $\mathcal{N}(0, A^2)$ restricted to $\mathbb{R}^+ = (0, \infty)$
- Observations are independent given the group means and σ
- Fix $m_0, s_0, A_\tau, A_\sigma$ using subject knowledge (e.g. prior predictive distribution)



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A half-normal prior for a standard deviation

A simple way to restrict τ to positive values is

$$Z \sim \mathcal{N}(0, A_\tau^2), \qquad \tau = |Z|.$$

This defines $\tau \sim \text{HalfNormal}(0, A_\tau)$ on $\mathbb{R}^+ = (0, \infty)$, with

$$\Pr(\tau < 1.96 A_\tau) \approx 0.95.$$

- The 0 denotes the mean of the underlying normal Z, not of τ
- A_τ has the same units as the outcome
- Example: to put 95% prior probability below a between-machine SD of 20 quality units, choose $A_\tau \approx 20/1.96 = 10.2$
- Choose A_σ separately for variation within a machine
- These scales are illustrative, not universal defaults



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The prior on variation affects pooling

- Small τ: machine means lie close to μ, encouraging stronger pooling
- Large τ: machine means can differ more, allowing less pooling
- With few groups, the data may give little information about τ, so its prior can matter substantially
- Check prior predictive outcomes and compare plausible prior scales
- A uniform prior on τ, on τ^2, or on $\log \tau$ gives different assumptions; "flat" does not mean neutral

BDA3, Section 5.7 discusses priors for hierarchical variation. The half-normal specification here is one modelling choice.



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Factory: observations and means

- In the common-variance model, observations are independent given θ_j and σ:

$$y_{ij} \mid \theta_j, \sigma \sim \mathcal{N}(\theta_j, \sigma^2), \qquad i = 1, \ldots, n_j.$$

- Machine j's sample mean and its sampling variance are

$$\bar{y}_{.j} = \frac{1}{n_j} \sum_{i=1}^{n_j} y_{ij},$$

$$\text{Var}(\bar{y}_{.j} \mid \theta_j, \sigma) = \frac{\sigma^2}{n_j} = s_j^2.$$

SD of observations: σ; SE of the mean: $s_j = \sigma/\sqrt{n_j}$.



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- Machine j's sample mean and its sampling variance are

$$\bar{y}_{.j} = \frac{1}{n_j} \sum_{i=1}^{n_j} y_{ij},$$

$$\text{Var}(\bar{y}_{.j} \mid \theta_j, \sigma) = \frac{\sigma^2}{n_j} = s_j^2.$$

  SD of observations: σ; SE of the mean: $s_j = \sigma/\sqrt{n_j}$.

- Thus $\bar{y}_{.j} \mid \theta_j, \sigma \sim \mathcal{N}(\theta_j, s_j^2)$. We condition on σ here; full Bayes also learns σ from the data.



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Normal shrinkage: the weights

For fixed μ, $\tau > 0$ and $\sigma > 0$, let $s_j = \sigma/\sqrt{n_j}$.

$$\mathrm{E}(\theta_j \mid \mu, \tau, \sigma, y) = w_j \bar{y}_{.j} + (1 - w_j)\mu, \qquad w_j = \frac{\tau^2}{\tau^2 + s_j^2}.$$

- A precise machine mean (small s_j) receives more weight
- Small between-machine variation τ gives more weight to μ
- As $\tau \to 0$, the conditional mean approaches μ; as $\tau \to \infty$, it approaches $\bar{y}_{.j}$
- In full Bayes, average over $p(\mu, \tau, \sigma \mid y)$
- Strong pooling makes the machine means similar; their common value can still be uncertain because μ is unknown



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Normal shrinkage: a numerical example

Suppose machine 1 has $\bar{y}_{.1} = 28$ from $n_1 = 4$ observations. For illustration, **fix** $\mu = 8$, $\tau = 5$ and $\sigma = 30$, so $s_1 = 30/\sqrt{4} = 15$.

$$w_1 = \frac{5^2}{5^2 + 15^2} = \frac{25}{250} = 0.10,$$
$$m_1 = 0.10 \cdot 28 + 0.90 \cdot 8 = 10,$$
$$v_1 = \left(s_1^{-2} + \tau^{-2}\right)^{-1} = \left(\frac{1}{225} + \frac{1}{25}\right)^{-1} = 22.5.$$

Posterior precision = data precision + prior precision.

The conditional posterior is $\theta_1 \mid \mu, \tau, \sigma, y \sim \mathcal{N}(m_1, v_1)$: mean 10, SD $\sqrt{22.5} \approx 4.74$.

- The noisy machine mean receives 10% of the weight
- The full factory posterior also averages over μ, τ, σ



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Pair discussion: precision and shrinkage

Discuss in pairs (1–2 minutes).

Two machines have the same sample mean, different from μ. They share a within-machine SD σ, but different sample sizes give different standard errors of their means:

$$s_1 = 5, \qquad s_2 = 20.$$

Hold μ, $\tau > 0$ and $\sigma > 0$ fixed.

Which machine's estimate is shrunk more toward μ? Why?



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Pair discussion: precision and shrinkage

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$$s_1 = 5, \qquad s_2 = 20.$$

Hold μ, $\tau > 0$ and $\sigma > 0$ fixed.

Which machine's estimate is shrunk more toward μ? Why?

Answer: Machine 2. Its larger standard error means less precise evidence, so its sample mean receives less weight and μ receives more weight.



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Prediction for an existing machine

In the common-variance model, distinguish the mean θ_j from a future observation $\tilde{y}_j$:

$$p(\tilde{y}_j \mid y) = \int\int p(\tilde{y}_j \mid \theta_j, \sigma) p(\theta_j, \sigma \mid y)\, d\theta_j\, d\sigma.$$

For each posterior draw s:

1. Take $(\theta_j^{(s)}, \sigma^{(s)})$ from the **same joint posterior draw**
2. Draw $\tilde{y}_j^{(s)} \sim \mathcal{N}(\theta_j^{(s)}, (\sigma^{(s)})^2)$

- The $\theta_j^{(s)}$ summarize uncertainty about the machine's mean
- The $\tilde{y}_j^{(s)}$ also include variation in a new observation



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Prediction for a new similar machine

With no observations yet from the new machine,

$$p(\theta_{\text{new}} \mid y) = \int\int p(\theta_{\text{new}} \mid \mu, \tau) p(\mu, \tau \mid y)\, d\mu\, d\tau,$$

$$p(\tilde{y}_{\text{new}} \mid y) = \int\int p(\tilde{y}_{\text{new}} \mid \theta_{\text{new}}, \sigma) p(\theta_{\text{new}}, \sigma \mid y)\, d\theta_{\text{new}}\, d\sigma.$$

For each posterior draw s:

1. Take $(\mu^{(s)}, \tau^{(s)}, \sigma^{(s)})$ jointly from $p(\mu, \tau, \sigma \mid y)$
2. Draw the new mean: $\theta_{\text{new}}^{(s)} \sim \mathcal{N}(\mu^{(s)}, (\tau^{(s)})^2)$
3. For a new observation, also draw $\tilde{y}_{\text{new}}^{(s)} \sim \mathcal{N}(\theta_{\text{new}}^{(s)}, (\sigma^{(s)})^2)$

Predicting a new mean stops at step 2; predicting an observation includes step 3.



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Pair discussion: a new machine

Discuss in pairs (1–2 minutes).

We have data from six machines and want to predict for a new, similar machine 7.

- Its mean quality, θ_7
- The quality of its next item, $\tilde{y}_7$

How do the simulation steps differ? What additional uncertainty enters the second prediction?



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Pair discussion: a new machine

Discuss in pairs (1–2 minutes).

We have data from six machines and want to predict for a new, similar machine 7.

- Its mean quality, θ_7
- The quality of its next item, $\tilde{y}_7$

How do the simulation steps differ? What additional uncertainty enters the second prediction?

Answer: Both start by drawing a new machine mean using posterior draws of the population parameters. Predicting an item also adds within-machine variation:

$$\tilde{y}_7^{(s)} \sim \mathcal{N}(\theta_7^{(s)}, (\sigma^{(s)})^2).$$



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Section 2

Exchangeability



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Independence and exchangeability

1. **Independence:** knowing one outcome does not change the distribution of another.
2. **Exchangeability:** swapping the labels or positions of observations leaves their joint distribution unchanged.



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Exchangeability

- Motivates a symmetric joint model
  - for observations whose labels carry no relevant information
  - for group parameters that we treat symmetrically a priori
- Less restrictive than **IID** (independent and identically distributed)
- IID implies exchangeability, but exchangeability need not imply independence
- Independence alone is not enough: the distributions may differ



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Exchangeability

- *Exchangeability*
  Random variables $\theta_1, \ldots, \theta_J$ (or $y_1, \ldots, y_J$) are exchangeable if the joint distribution p is invariant to the permutation of indices
  i.e

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_{\pi(1)}, \theta_{\pi(2)}, \theta_{\pi(3)})$$

  for any permutation π of the indices. E.g.

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$



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$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$

  Can we come up with a situation where this doesn't hold?



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$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$

  Can we come up with a situation where this doesn't hold?
  E.g. when we have a trend over indices (like in time series data)



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  for any permutation π of the indices. E.g.
  $$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$
  Can we come up with a situation where this doesn't hold?
  E.g. when we have a trend over indices (like in time series data)
- Exchangeability implies symmetry:
  If there is no information which can be used *a priori* to separate θ_j from each other, we can assume exchangeability. ("Ignorance implies exchangeability")



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Exchangeability: de Finetti's theorem

- **Bernoulli case:** let $(X_i)_{i=1}^{\infty}$ be an infinite exchangeable sequence with $X_i \in \{0, 1\}$
- There is a random $\theta \in [0, 1]$ such that the X_i are conditionally IID:

$$X_i \mid \theta \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$$

- If θ has a density $p(\theta)$, then for every finite J,

$$p(x_1, \ldots, x_J) = \int_0^1 \left[\prod_{i=1}^{J} \theta^{x_i} (1-\theta)^{1-x_i} \right] p(\theta)\, d\theta$$

- The theorem does not choose the mixing distribution: a Beta prior is a modelling choice, not a consequence



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De Finetti's takeaway

- **Usually, we start with a Bayesian model:** an unknown coin bias θ, a prior for θ, and independent flips given θ.
- **De Finetti starts with symmetry:** the order of the flips carries no information.
- For an **infinite exchangeable sequence of binary outcomes**, our beliefs can be represented **as if** there were a shared unknown success probability θ.
- Given θ, the outcomes are independent; a prior distribution describes our uncertainty about θ. The outcomes need not be independent marginally.

We do not need to assume an objectively existing chance first: exchangeability justifies this familiar Bayesian representation.



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Pair discussion: learning about the coin

Discuss in pairs (1–2 minutes).

The coin's bias has prior $\theta \sim \text{Beta}(1, 1)$, and the flips are independent given θ.

Why does observing heads change our prediction for the next flip? Does this contradict conditional independence?



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Why does observing heads change our prediction for the next flip? Does this contradict conditional independence?

Answer: We learn about the shared unknown bias. After observing heads, $\theta \mid \text{heads} \sim \text{Beta}(2, 1)$, changing the predictive probability from $1/2$ to $2/3$.

The coin has not changed. Independence holds when θ is fixed; averaging over an uncertain θ creates dependence.



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Coin flips: exchangeable but dependent

Let $\theta \sim \text{Beta}(1, 1)$ and, given θ, let the flips be IID Bernoulli(θ). Write H_i for heads on flip i.

$$\Pr(H_1) = \mathrm{E}(\theta) = \frac{1}{2}.$$



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Let $\theta \sim \text{Beta}(1,1)$ and, given θ, let the flips be IID Bernoulli(θ). Write H_i for heads on flip i.

$$\Pr(H_1) = \mathrm{E}(\theta) = \frac{1}{2}.$$

After observing H_1,

$$\theta \mid H_1 \sim \text{Beta}(2,1), \qquad \Pr(H_2 \mid H_1) = \mathrm{E}(\theta \mid H_1) = \frac{2}{3}.$$

- Marginally $\Pr(H_2) = 1/2$, so the flips are dependent
- Their joint distribution remains unchanged when we swap the flips
- Observing heads updates the shared unknown bias; one toss does not physically affect the next



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Exchangeability

- Exchangeability does not require identical experiment probabilities
- Suppose experiments came from two laboratories with different conditions, but their laboratory memberships are unknown
- If we treat these unknown memberships symmetrically across experiments, we can model the probabilities θ_j as **exchangeable a priori**
- A model with unknown laboratory membership can represent clustering among experiments from the same laboratory
- This concerns θ_j, not the counts y_j: different sample sizes n_j can make the counts non-exchangeable



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Exchangeability and additional information

- Bioassay: doses x_j and group sizes n_j are fixed design information; y_j counts deaths



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Exchangeability and additional information

- Bioassay: doses x_j and group sizes n_j are fixed design information; y_j counts deaths
- Given α, β, the counts are independent, but generally not identically distributed:

$$y_j \mid \alpha, \beta, n_j, x_j \sim \text{Binomial}(n_j, \pi_j), \qquad j = 1, \ldots, J$$

where $\text{logit}(\pi_j) = \alpha + \beta x_j$.

$$p(\alpha, \beta \mid y, n, x) \propto p(\alpha, \beta) \prod_{j=1}^{J} p(y_j \mid \alpha, \beta, n_j, x_j)$$



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$$p(\alpha, \beta \mid y, n, x) \propto p(\alpha, \beta) \prod_{j=1}^{J} p(y_j \mid \alpha, \beta, n_j, x_j)$$

- Reordering complete records (x_j, n_j, y_j) leaves the likelihood unchanged; swapping only counts between doses generally changes it



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Hierarchical exchangeability

- Rat example: experiments $j = 1, \ldots, J$, each with n_j rats



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Hierarchical exchangeability

- Rat example: experiments $j = 1, \ldots, J$, each with n_j rats
- Within experiment j, outcomes are IID Bernoulli(θ_j) given θ_j; they need not be exchangeable across experiments



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- Rat example: experiments $j = 1, \ldots, J$, each with n_j rats
- Within experiment j, outcomes are IID Bernoulli(θ_j) given θ_j; they need not be exchangeable across experiments
- The group probabilities θ_j are modelled exchangeably a priori



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Hierarchical exchangeability

- Rat example: experiments $j = 1, \ldots, J$, each with n_j rats
- Within experiment j, outcomes are IID Bernoulli(θ_j) given θ_j; they need not be exchangeable across experiments
- The group probabilities θ_j are modelled exchangeably a priori
- Counts $y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j)$ need not be exchangeable when n_j differs
- This assumes comparable experiments; grouping alone does not justify a shared population distribution



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Partial or conditional exchangeability

- **Conditional exchangeability**
  - Responses with the same covariates may be modelled exchangeably; different covariates may imply different distributions
  - For fixed covariates, use a conditional model $p(y \mid x)$; adding x_i does not make (x_i, y_i) exchangeable



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Partial or conditional exchangeability

- **Conditional exchangeability**
  - Responses with the same covariates may be modelled exchangeably; different covariates may imply different distributions
  - For fixed covariates, use a conditional model $p(y \mid x)$; adding x_i does not make (x_i, y_i) exchangeable
- **Partial exchangeability**
  - Symmetry within specified groups, not under arbitrary exchanges across groups
  - The models here additionally assume exchangeable group parameters, possibly given group-level covariates



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Section 3

Computational aspects



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Funnel geometry: prior and posterior

- Consider the **prior hierarchy**, before observing data:

$$\tau \sim p(\tau), \quad \tau > 0, \qquad \theta_j \mid \tau \sim \mathcal{N}(0, \tau) \quad \text{(SD form)}$$

- Small τ: group effects lie near 0; large τ: group effects are more spread out
- If plausible τ values range from near zero to large values, the joint prior has a narrow neck and a broad region: a funnel



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Funnel geometry: prior and posterior

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- Small τ: group effects lie near 0; large τ: group effects are more spread out
- If plausible τ values range from near zero to large values, the joint prior has a narrow neck and a broad region: a funnel
- With weak information about group effects, the **posterior can retain** this funnel geometry
- Not every hierarchical posterior is funnel-shaped: the likelihood also matters

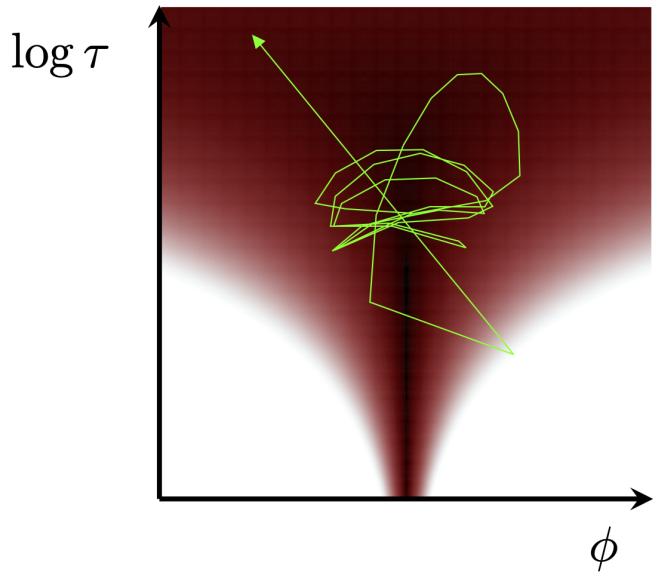


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Why is a funnel problematic?

- Changing curvature makes it hard for HMC to explore both the narrow and wide regions efficiently
- Post-warmup divergences warn of potentially unreliable exploration and biased posterior estimates
- Good $\widehat{R}$ and ESS do not cancel divergence warnings



Betancourt (2020) Stan diagnostics HMC demo

In this figure, ϕ denotes one group effect θ_j.



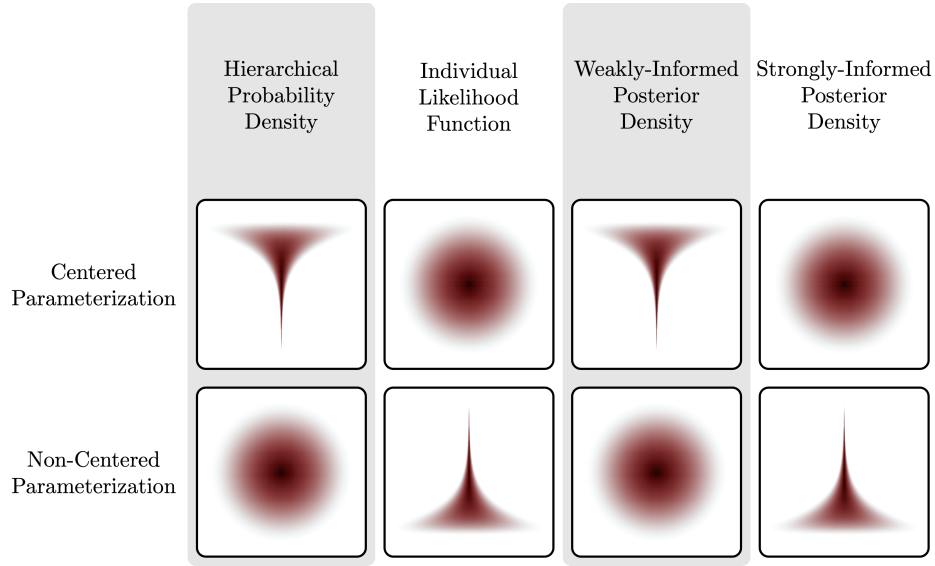
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Handling the funnel

- Raising `adapt_delta` targets smaller steps: it can help, but does not guarantee reliable inference
- **Centered:** often better with strong group information

$$\theta_j \sim N(\mu, \tau) \quad \text{(SD form)}$$



Betancourt (2020) Stan: parameterization guidance



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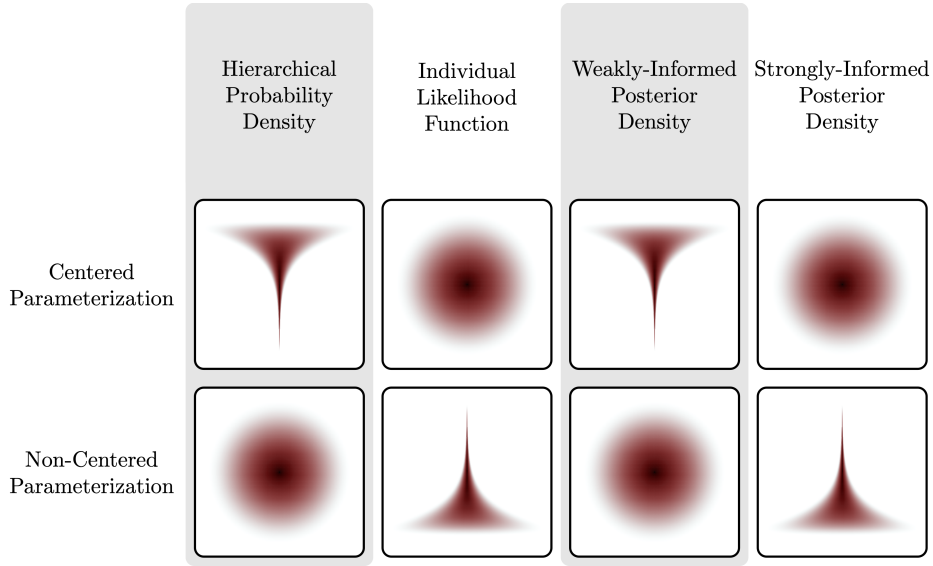
Handling the funnel

- Raising `adapt_delta` targets smaller steps: it can help, but does not guarantee reliable inference
- **Centered:** often better with strong group information

$$\theta_j \sim N(\mu, \tau) \quad \text{(SD form)}$$

- **Non-centered:** often helps weakly informed group effects

$$\eta_j \sim N(0, 1), \qquad \theta_j = \mu + \tau\eta_j$$



Betancourt (2020) Stan: parameterization guidance



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Why non-centering preserves the model

Start with $\eta_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, independent of the hyperparameters a priori, and set

$$\theta_j = \mu + \tau\eta_j, \qquad \tau > 0.$$

Conditional on μ, τ, an affine transformation of a normal variable is normal, with

$$\begin{aligned} \mathsf{E}(\theta_j \mid \mu, \tau) &= \mu + \tau\,\mathsf{E}(\eta_j) = \mu, \\ \mathsf{Var}(\theta_j \mid \mu, \tau) &= \tau^2\,\mathsf{Var}(\eta_j) = \tau^2. \end{aligned}$$

Thus the θ_j are conditionally independent normals with mean μ and SD τ, as in the centered hierarchy.

Keep the same hyperpriors and likelihood, evaluated at $\theta_j = \mu + \tau\eta_j$. Only the sampling coordinates change.



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Section 4

Additional material

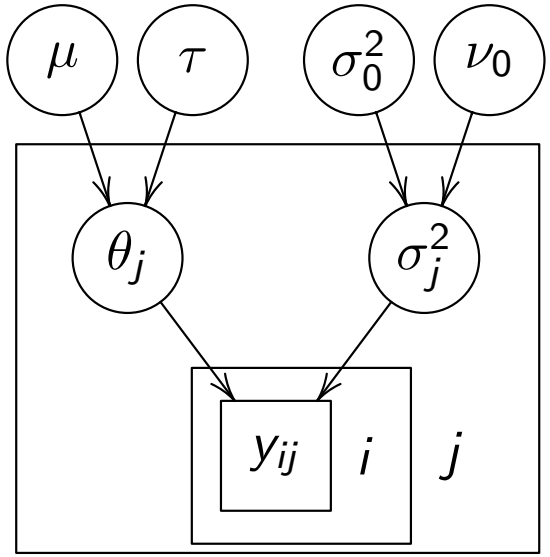


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Factory: separate variances

- Six machines with means θ_j and own variances σ_j^2



$$y_{ij} \mid \theta_j, \sigma_j^2 \sim \mathcal{N}(\theta_j, \sigma_j^2)$$
$$\theta_j \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2)$$
$$\sigma_j^2 \mid \nu_0, \sigma_0^2 \sim \text{Inv}{-}\chi^2(\nu_0, \sigma_0^2)$$

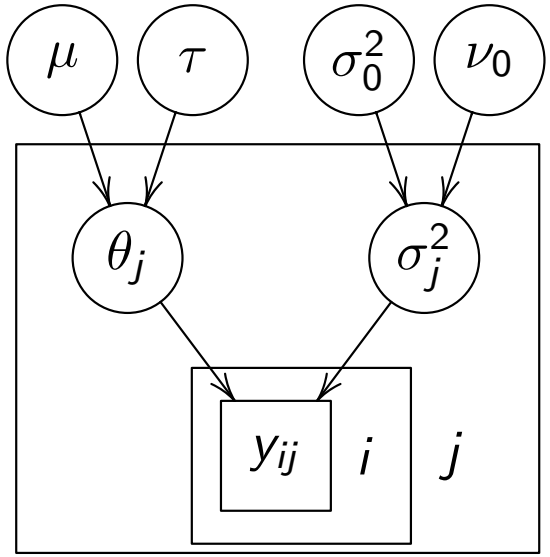


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- **Pair discussion (1–2 minutes):** What predictive distributions can we have?

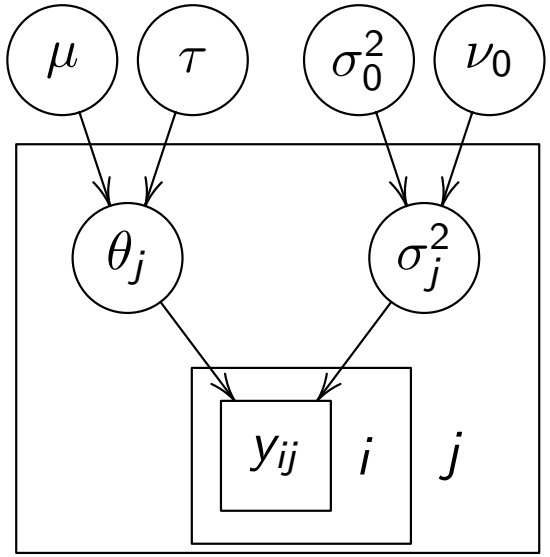


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- **Pair discussion (1–2 minutes):** What predictive distributions can we have?
- Observations from existing or new similar machines



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Eight schools: historical SAT coaching study

- Historical example from BDA3, Section 5.5
  - randomized coaching experiments in eight high schools
  - question: does short-term coaching improve verbal test scores?
  - estimate a coaching effect for each school



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Eight schools: historical SAT coaching study

- Historical example from BDA3, Section 5.5
  - randomized coaching experiments in eight high schools
  - question: does short-term coaching improve verbal test scores?
  - estimate a coaching effect for each school
- SAT-V in this study (the historical verbal test)
  - standardized multiple-choice test
  - score range 200–800, with mean about 500 and SD about 100
  - pre-test PSAT scores in verbal (V) and mathematics (M) help adjust for students' initial performance



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Hierarchical normal model: 8 schools

- Effectiveness of the SAT coaching
  - students took pre-tests PSAT-M and PSAT-V
  - some students received coaching
  - a regression in each school adjusts for pre-test scores and estimates the coaching effect y_j and its standard error σ_j
  - use a normal approximation for y_j, treating the estimated sampling variances σ_j^2 as known
  - y_j is an adjusted effect estimate, not a sample mean; σ_j is not the SD of individual students' scores



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  - y_j is an adjusted effect estimate, not a sample mean; σ_j is not the SD of individual students' scores
- Data:

School	A	B	C	D	E	F	G	H
y_j	28	8	-3	7	-1	1	18	12
σ_j	15	10	16	11	9	11	10	18



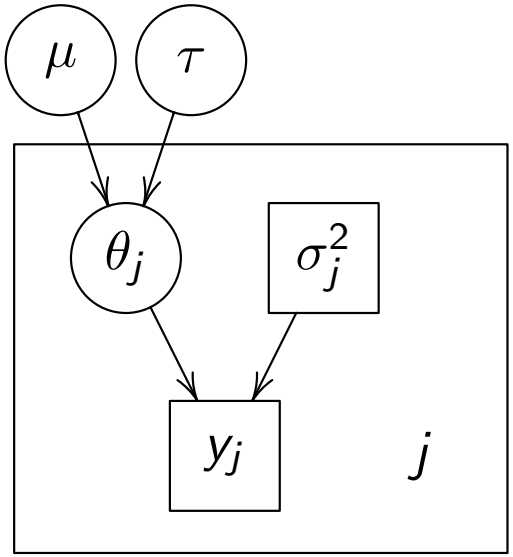
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Hierarchical normal model for school effects

$$\theta_j | \mu, \tau \sim \mathcal{N}(\mu, \tau)$$

$$y_j | \theta_j \sim \mathcal{N}(\theta_j, \sigma_j^2)$$



The observed y_j is a school effect estimate; σ_j is its reported standard error, treated as known.

Here $\mathcal{N}(\mu, \tau)$ is written in mean–SD form; τ is the between-school SD.



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Normal shrinkage: completing the square

For fixed μ and $\tau > 0$, with known sampling SD $\sigma_j > 0$,

$$y_j \mid \theta_j \sim \mathcal{N}(\theta_j, \sigma_j^2), \qquad \theta_j \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2).$$

Here $\mathcal{N}$ uses variance as its second argument.

$$\begin{aligned} p(\theta_j \mid \mu, \tau, y) &\propto \exp\left[-\frac{1}{2}\left\{\frac{(y_j - \theta_j)^2}{\sigma_j^2} + \frac{(\theta_j - \mu)^2}{\tau^2}\right\}\right] \\ &\propto \exp\left[-\frac{(\theta_j - m_j)^2}{2v_j}\right], \end{aligned}$$

where completing the square gives

$$v_j = \left(\frac{1}{\sigma_j^2} + \frac{1}{\tau^2}\right)^{-1}, \qquad m_j = v_j\left(\frac{y_j}{\sigma_j^2} + \frac{\mu}{\tau^2}\right).$$

Thus $\theta_j \mid \mu, \tau, y \sim \mathcal{N}(m_j, v_j)$.

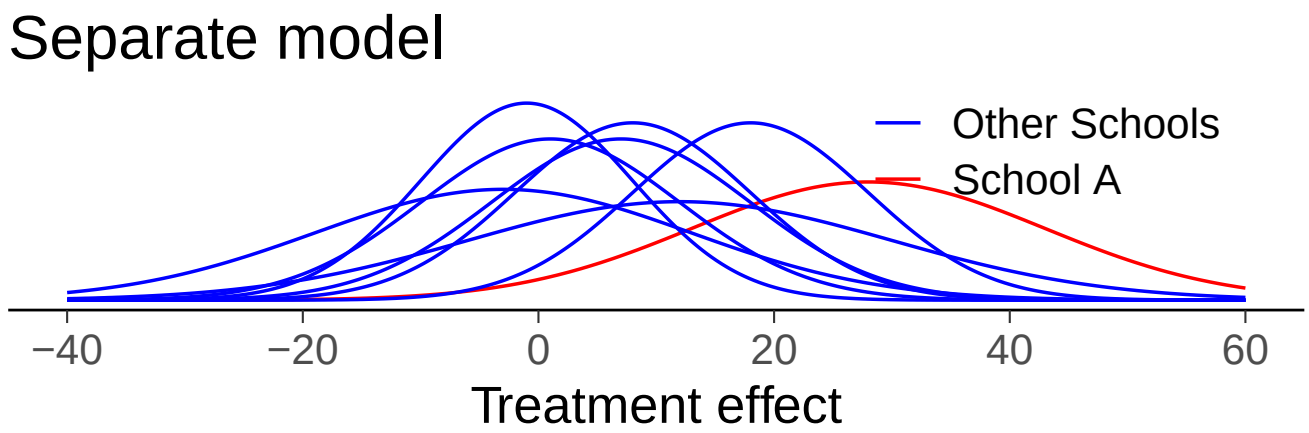


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Eight schools: pooling assumptions

How does the posterior for School A (red) change?



No pooling: each school's effect uses its own data. School A's estimate is high but uncertain.

The hierarchical plots follow BDA3, Section 5.5, with flat independent priors on μ and $\tau > 0$, rather than the illustrative half-normal prior in the factory example.

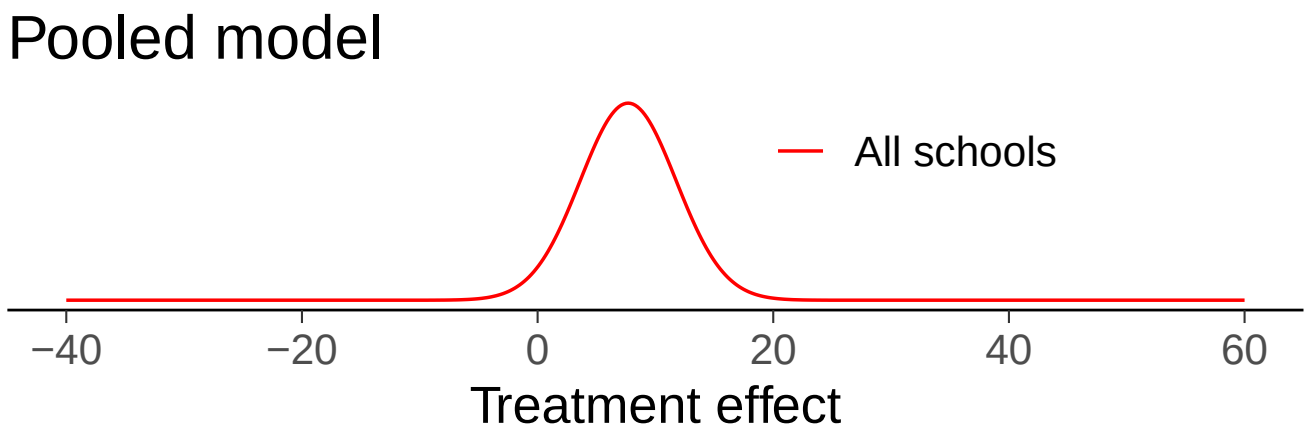


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Eight schools: pooling assumptions

How does the posterior for School A (red) change?



Complete pooling: all schools share a common effect. School A's posterior is identical to the others.

The hierarchical plots follow BDA3, Section 5.5, with flat independent priors on μ and $\tau > 0$, rather than the illustrative half-normal prior in the factory example.

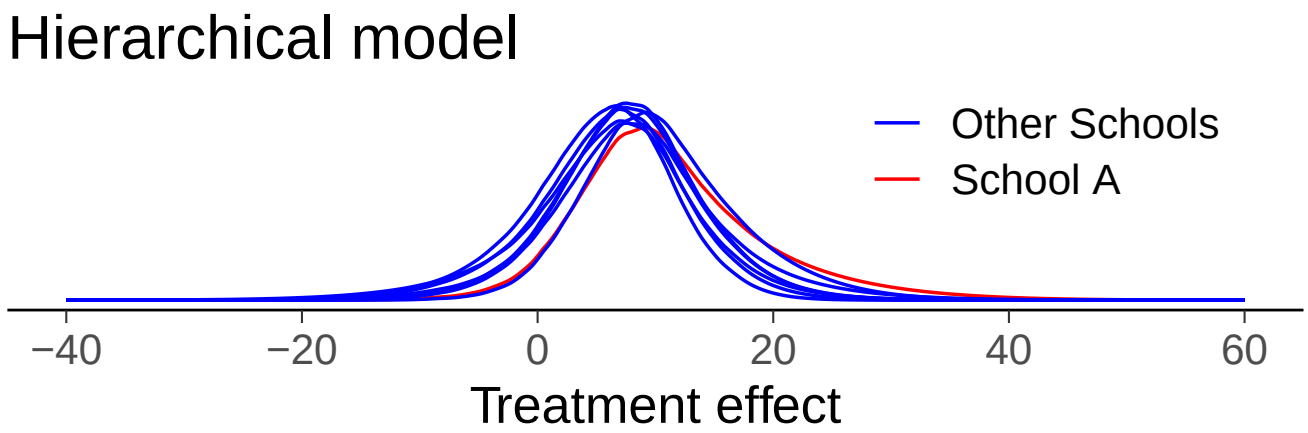


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Eight schools: pooling assumptions

How does the posterior for School A (red) change?



Partial pooling: the marginal posteriors $p(\theta_j \mid y)$ share information while allowing different effects.

The hierarchical plots follow BDA3, Section 5.5, with flat independent priors on μ and $\tau > 0$, rather than the illustrative half-normal prior in the factory example.

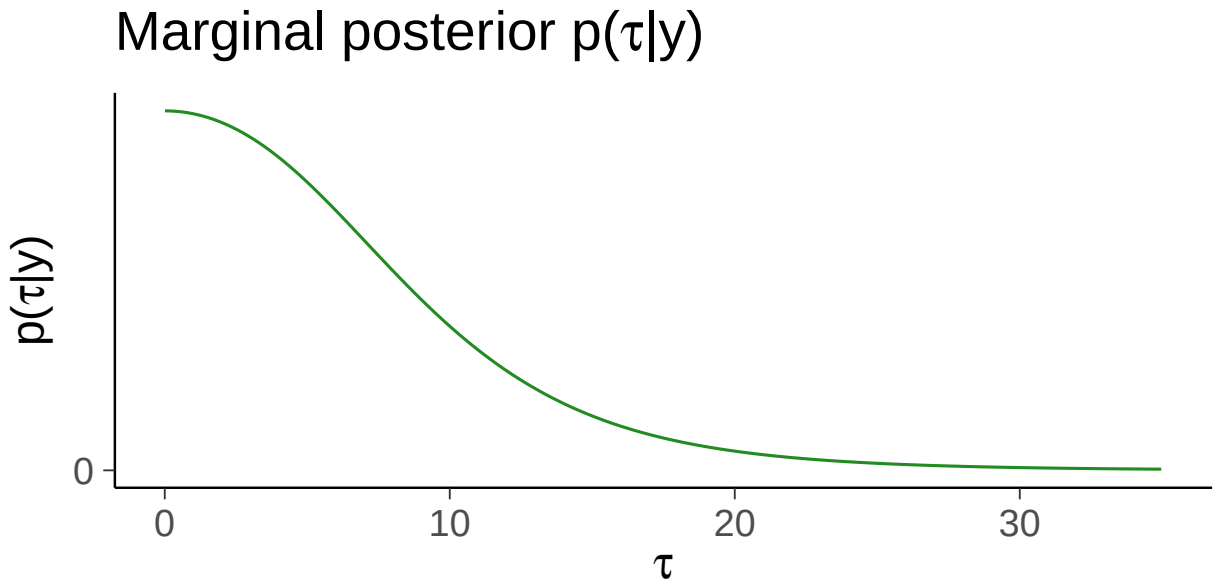


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Eight schools: uncertainty about variation

Does the peak near zero establish that all effects are equal?



- This is $p(\tau \mid y)$, with μ and the school effects integrated out
- A range of between-school SDs remains plausible
- Full Bayes averages over this distribution; fixing τ at its mode would impose much stronger pooling

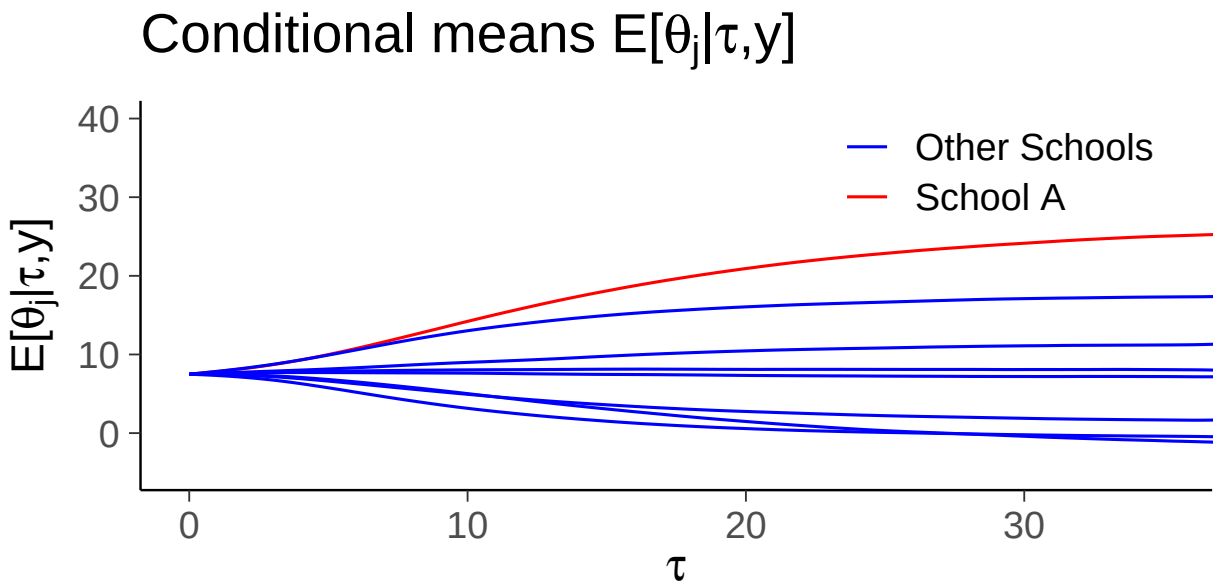


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Eight schools: conditional posterior means

What happens to the school estimates as τ changes?



- Curves show $\mathrm{E}(\theta_j \mid \tau, y)$: τ is fixed and μ is integrated out
- Small τ: means come together near the pooled estimate, 7.7
- Large τ: means approach the separate estimates y_j

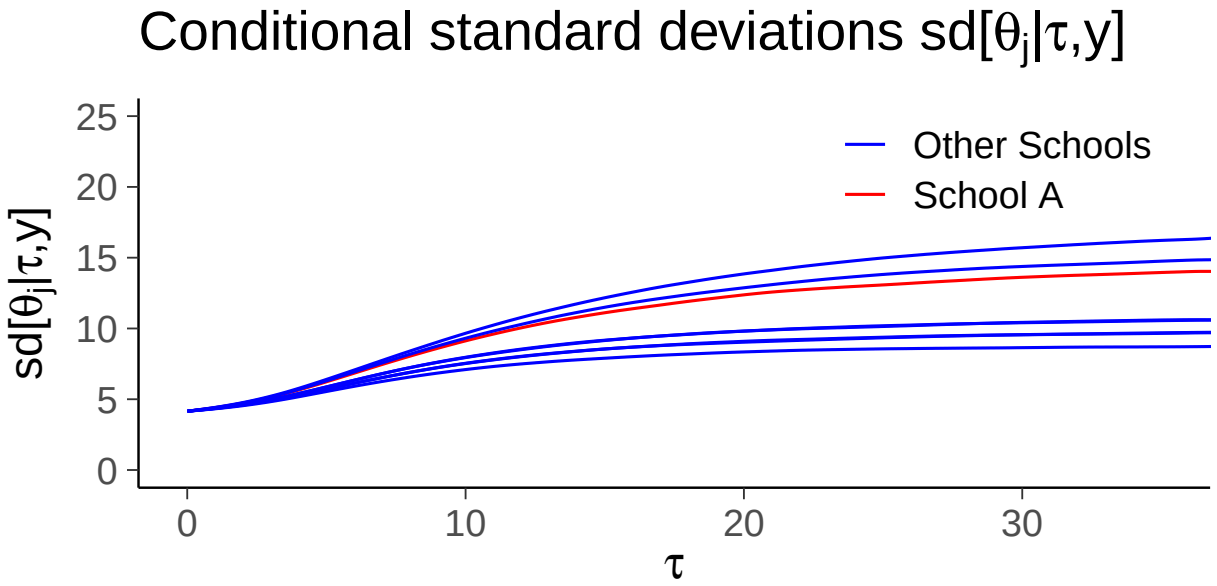


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Eight schools: uncertainty given τ

Does small τ make the common effect precisely known?



- Curves show $\text{sd}(\theta_j \mid \tau, y)$, with uncertainty in μ included
- As $\tau \to 0$, effects coincide, but their common value is uncertain:

$$\text{sd}(\theta_j \mid \tau, y) \longrightarrow \left(\sum_{k=1}^{J} \frac{1}{\sigma_k^2} \right)^{-1/2} \approx 4.1.$$